Distribuciones Continuas

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Momento de orden n (respecto al origen, μ'_n): $E[X^n]$

Momento central de orden (respecto a la media, μ, μ'_n):$E[(X-\mu)^n]$

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x)dx$$

Funcion Generadora de Momento **(FGM)**

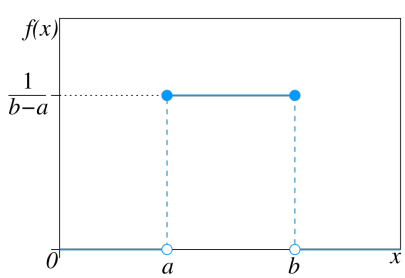
$$E[X^n] = \frac{d^n}{dt^n} M_X(t)\Big|_{t=0} \qquad con \qquad M_X(t) = E[e^{tX}]$$

Media: $E[X] = \int_a^b x f(x)dx$ **Varianza:** $V[X] = \sigma^2 = E[(X-\mu)^2] = E[X^2] - (E[X])^2$

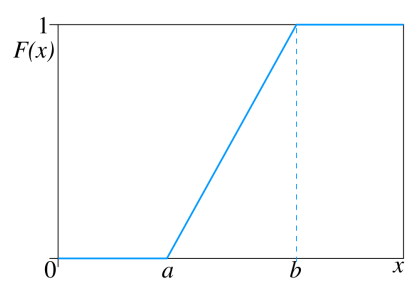
Asimetria: $\gamma_1 = \dfrac{E[(X-\mu)^3]}{\sigma^3}$ **Curtosis:** $\gamma_2 = \dfrac{E[(X-\mu)^4]}{\sigma^4}$

Uniforme $\mathcal{U}(a,b)$ $a < b$

Función de densidad $f(x) = \dfrac{1}{b-a}$



Función acumulativa $F(x) = \dfrac{x-a}{b-a}$

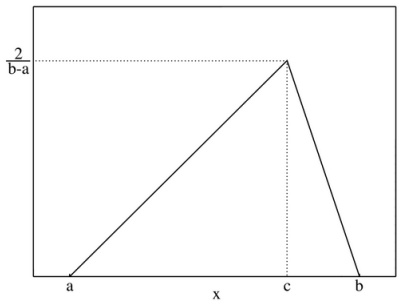


Media $E[x] = \dfrac{a+b}{2}$ Varianza $V[x] = \dfrac{(b-a)^2}{12}$ $FGM = \dfrac{e^{tb} - e^{ta}}{t(b-a)}$ Curtosis $\gamma_2 = \dfrac{-6}{5}$

Triangular $\quad Tri(a,b,c) \quad a < c < b$

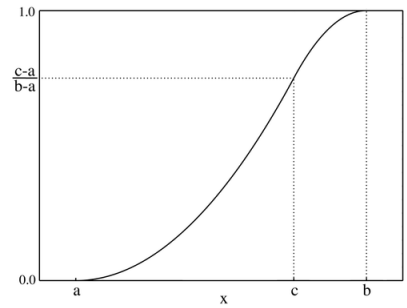
Función de densidad

$$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \le x \le c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \le x \le b \\ 0 & \text{en otro caso} \end{cases}$$



Función acumulativa

$$F(x) \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \le x \le c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \le x \le b \\ 1 & x > b \end{cases}$$

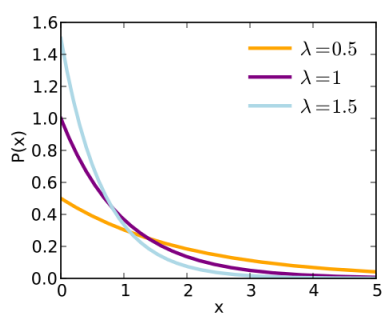


$$E[x] = \frac{a+b+c}{3} \qquad V[x] = \frac{a^2+b^2+c^2-ab-ac-bc}{18} \qquad FGM = 2\frac{(b-c)e^{at}-(b-a)e^{ct}+(c-a)e^{bt}}{(b-a)(c-a)(b-c)t^2} \qquad \gamma_2 = -\frac{3}{5}$$

Exponencial $\quad Exp(\lambda) \quad \lambda > 0 \quad x \in [0, \infty)$

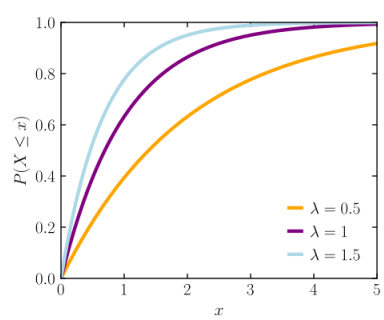
Función de densidad

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \ge 0 \\ 0 & x < 0 \end{cases}$$



Función acumulativa

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x \ge 0 \\ 0 & x < 0 \end{cases}$$

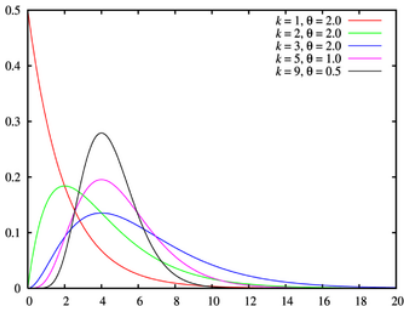


$$E[X] = \frac{1}{\lambda} \qquad V[X] = \frac{1}{\lambda^2} \qquad \gamma_1 = 2 \qquad \gamma_2 = 9 \qquad FGM = \left(1 - \frac{t}{\lambda}\right)^{-1}$$

m-Erlang $\quad Erlang(\lambda, n) \quad \lambda > 0 \quad n \in \mathbb{Z}^+ \quad x \in [0, \infty)$

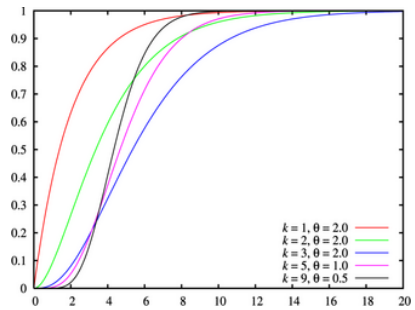
Función de densidad

$$f(x) = \begin{cases} \frac{\lambda(\lambda x)^{n-1} e^{-\lambda x}}{(n-1)!} & x \geq 0 \\ 0 & x < 0 \end{cases}$$



Función acumulativa

$$F(x) = 1 - \sum_{k=0}^{n-1} \frac{(\lambda x)^k e^{-\lambda x}}{k!}$$

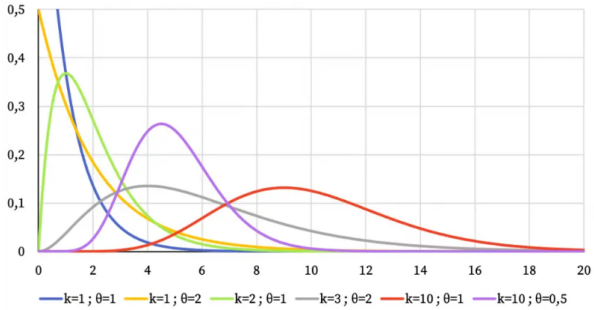


$$E[X] = \frac{n}{\lambda} \qquad V[X] = \frac{n}{\lambda^2} \qquad \gamma_1 = \frac{2}{\sqrt{n}} \qquad \gamma_2 = \frac{6}{n} \qquad FGM = \left(\frac{\lambda}{\lambda - t}\right)^n \qquad t < \lambda$$

Gamma $\quad Gamma(\alpha, \lambda) \quad \alpha, \lambda > 0$

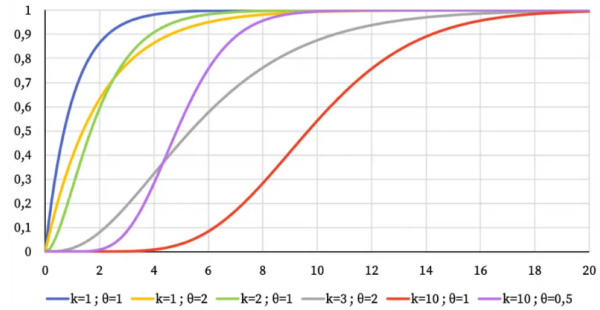
Función de densidad

$$f(x) = \frac{\lambda(\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$$



Función acumulativa

$$F(x) = \frac{1}{\Gamma(\alpha)} \gamma(\alpha, \lambda x)$$



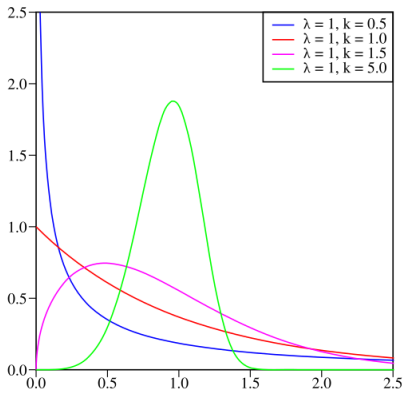
$$E[x] = \frac{\alpha}{\lambda} \qquad V[x] = \frac{\alpha}{\lambda^2} \qquad FGM = \left(\frac{\lambda}{\lambda - t}\right)^\alpha \qquad t < \lambda$$

Weibull $\quad We(\alpha,\beta) \quad \alpha,\beta>0 \quad x\in(0,\infty)$

Función de densidad

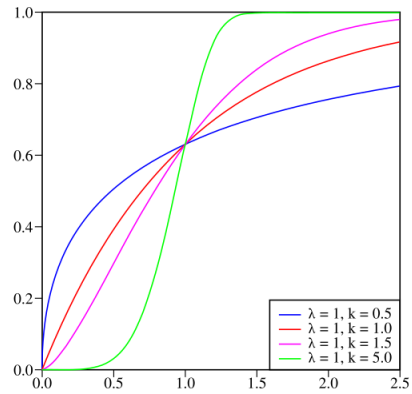
$$f(x)=\begin{cases}\alpha\beta^{-\alpha}x\alpha-1e^{-(x/\beta)^{\alpha}} & x>0\\ 0 & \text{otro caso}\end{cases}$$

$$f(x)=\lambda\alpha(\lambda x)^{\alpha-1}e^{(\lambda x)\alpha}$$



Función acumulativa

$$F(x)=\begin{cases}1-e^{-(\frac{x}{\beta})^{\alpha}} & \text{para } x>0\\ 0 & \text{otro caso}\end{cases}=1-e^{-(\lambda x)^{\alpha}}$$



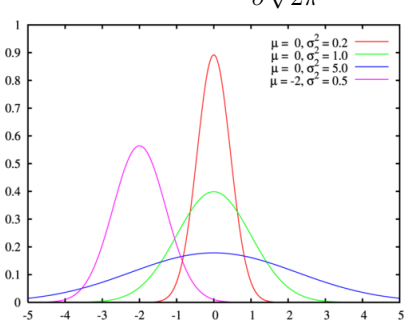
$$E[X]=\frac{\beta}{\alpha}\Gamma(\frac{1}{\alpha})=\frac{1}{\lambda}\Gamma(1+\frac{1}{\alpha}) \qquad V[X]=\frac{\beta^2}{\alpha}[2\Gamma(\frac{2}{\alpha})-\frac{1}{\alpha}[\Gamma(\frac{1}{\alpha})]^2]=\frac{1}{\lambda^2}[\Gamma(1+\frac{2}{\alpha})-\Gamma^2(1+\frac{1}{\alpha})]$$

$$\gamma_1=\frac{\Gamma(1+\frac{3}{\alpha})\frac{1}{\lambda^3}-3\mu\sigma^2-\mu^3}{\sigma^3} \qquad FGM=\sum_{n=0}^{\infty}\frac{t^n}{n!\lambda^n}\Gamma\left(1+\frac{n}{\alpha}\right) \qquad \alpha\geq 1$$

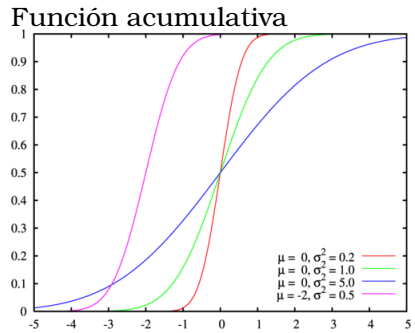
Normal $\quad N(\mu,\sigma^2) \quad \mu\in(-\infty,\infty) \quad \sigma>0 \quad x\in(-\infty,\infty)$

Función de densidad

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Función acumulativa

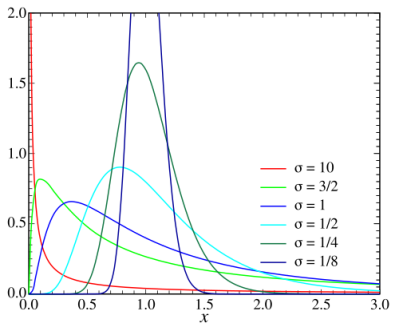


$$E[X]=\mu \qquad V[X]=\sigma^2 \qquad \gamma_1=0 \qquad \gamma_2=0 \qquad FGM=e^{\mu t+\frac{\sigma^2t^2}{2}}$$

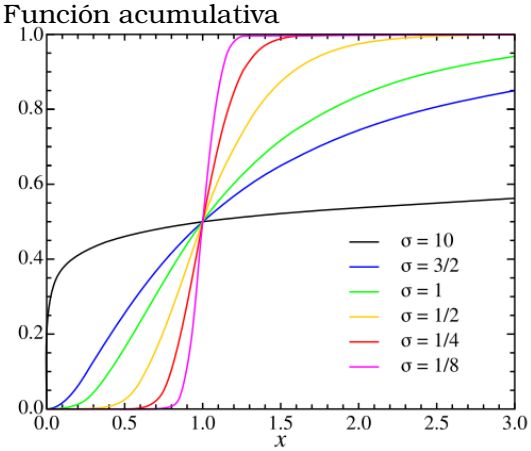
Log-Normal $LN(\mu, \sigma^2)$ $\sigma > 0$ $e^{\mu} > 0$ $x \in (-\infty, \infty)$

Función de densidad

$$f(x) = \begin{cases} \frac{1}{\sigma x \sqrt{2\pi}} e^{-\frac{(ln \quad x-\mu)^2}{2\sigma^2}} & x > 0 \\ 0 & \text{otro caso} \end{cases}$$



Función acumulativa



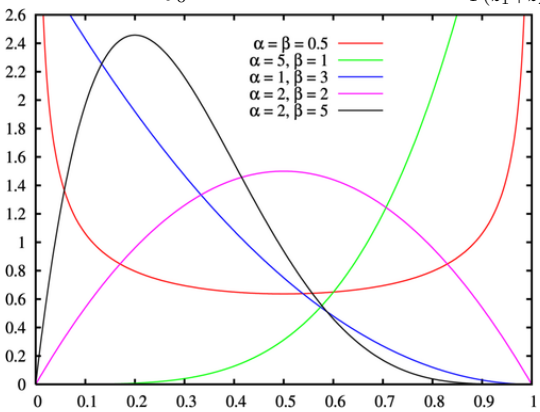
$$E[X] = e^{\mu+\sigma^2/2} \qquad V[X] = e^{\mu+\sigma^2/2}(e^{\sigma^2} - 1) \qquad \gamma_1 = (e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$$

Beta $\beta(\alpha, \beta)$ $\alpha, \beta > 0$

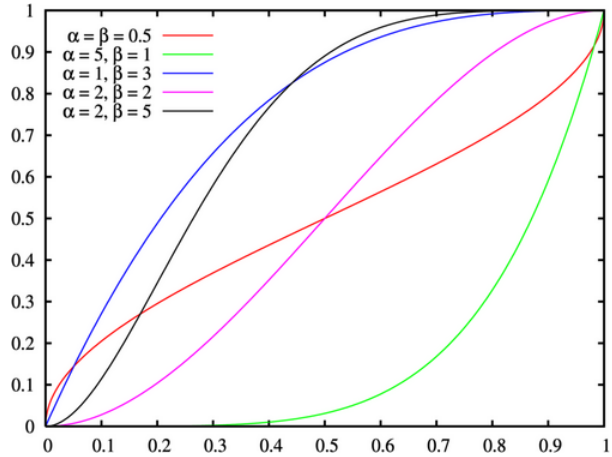
Función de densidad

$$f(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta 2)} = & \text{para } 0 \leq x \leq 1 \\ 0 & \text{en otro caso} \end{cases}$$

Con $B(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1} \cdot dt = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1+z_2)}$



Función acumulativa



$$E[X] = \frac{\alpha}{\alpha+\beta} \qquad V[X] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \qquad \gamma_1 = \frac{2(\beta - \alpha\sqrt{\alpha+\beta+1})}{(\alpha+\beta+2)\sqrt{\alpha\beta}} \qquad FGM = 1 + \sum_{n=1}^{\infty}\left(\prod_{r=0}^{n-1}\frac{\alpha+r}{\alpha+\beta+r}\right)\frac{t^n}{n!}$$

Pearson V $\quad PT5(\alpha,\beta) \quad \iff \frac{1}{X} \sim gamma(\alpha, \frac{1}{\beta})$

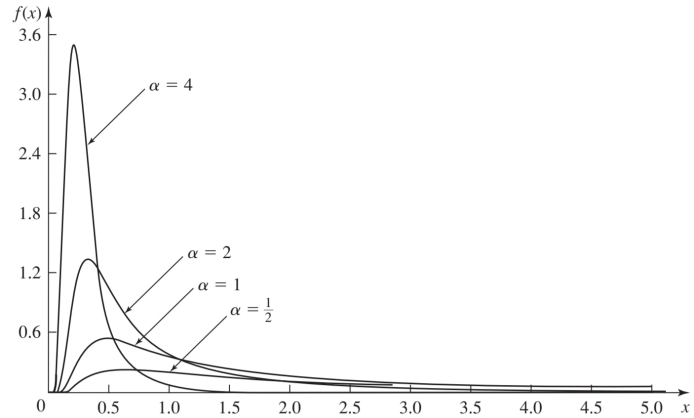
Función de densidad

$$f(x) = \begin{cases} \frac{x^{-(\alpha-1)}e^{-\frac{\beta}{x}}}{\beta^{-\alpha}\Gamma(\alpha)} & \text{para } x > 0 \\ 0 & \text{otra caso} \end{cases}$$

Función acumulativa

$$F(x) = \begin{cases} 1 - F_G(\frac{1}{x}) & \text{para } x > 0 \\ 0 & \text{otro caso} \end{cases}$$

Con F_G la función de distribución de $gamma(\alpha, \frac{1}{\beta})$



$$E[X] = \frac{\beta}{\alpha - 1} \qquad para \quad \alpha > 1 \qquad V[X] = \frac{\beta^2}{(\alpha-1)^2(\alpha-2)} \qquad para \quad \alpha > 2$$

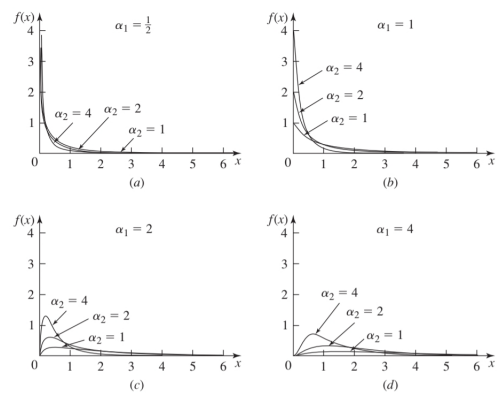
Pearson VI $\quad PT6(\alpha_1, \alpha_2, \beta) \qquad \alpha_1, \alpha_2, \beta > 0$

Función de densidad

$$f(x) = \begin{cases} \frac{(x/\beta)^{\alpha_1 - 1}}{\beta B(\alpha_1,\alpha_2)[1+(x/\beta)]^{\alpha_1+\alpha_2}} & \text{para } x > 0 \\ 0 & \text{otra caso} \end{cases}$$

Función acmulativa

$$F(x) = \begin{cases} F_B(\frac{x}{x+\beta}) & \text{para } x > 0 \\ 0 & \text{otro caso} \end{cases}$$



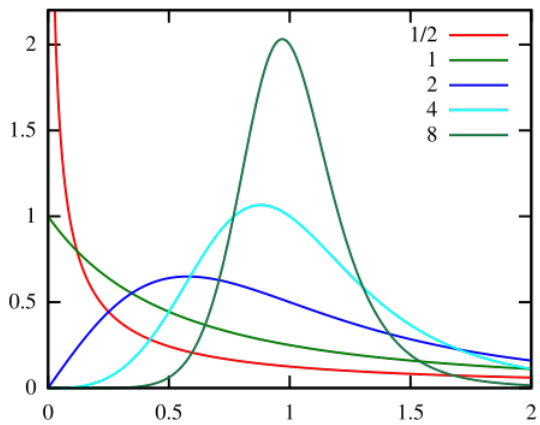
Con $F_B$la función de distribución beta $\beta(\alpha, \frac{1}{\beta})$

$$E[X] = \frac{\beta\alpha_1}{\alpha_2 - 1} \qquad para \quad \alpha_2 > 1 \qquad V[X] = \frac{\beta^2\alpha_1(\alpha_1+\alpha_2-1)}{(\alpha_2-1)^2(\alpha_2-2)} \qquad para \quad \alpha > 2$$

Log-Logística $LL(\alpha, \beta)$ $\alpha, \beta > 0$

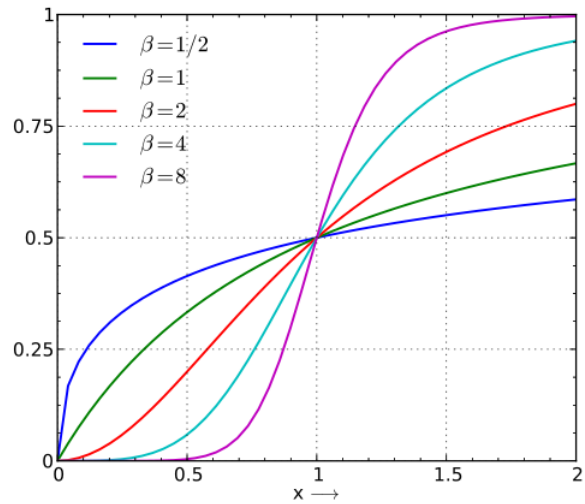
Función de densidad

$$f(x) = \begin{cases} \frac{\alpha(\frac{x}{\beta})^{\alpha-1}}{\beta[1+(\frac{x}{\beta})^{\alpha}]^2} & \text{para } x > 0 \\ 0 & \text{otro caso} \end{cases}$$



Función acumulativa

$$F(x) = \begin{cases} \frac{1}{1+(\frac{x}{\beta})^{-\alpha}} & \text{para } x > 0 \\ 0 & \text{otro caso} \end{cases}$$

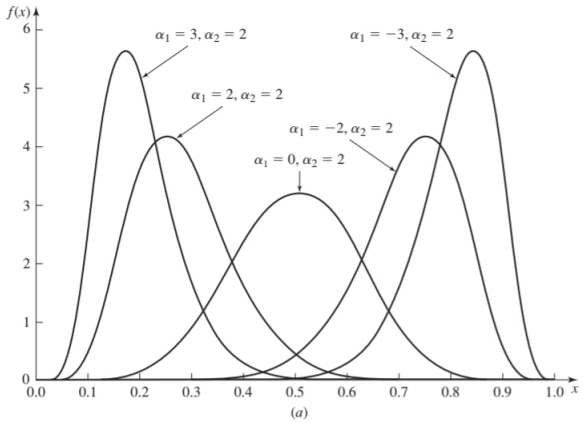


$$E[X] = \beta\theta \cdot cosecant(\theta) \qquad \alpha > 1 \qquad \theta = \frac{\pi}{\alpha} \qquad V[X] = \beta^2\theta[2cosecant(2\theta) - \theta[cosecant(\theta)]^2] \qquad \alpha > 2$$

Jhonson Bounded $JSB(\alpha_1, \alpha_2, a, b)$ $a \in (-\infty, \infty)$ $b > a$ $\alpha_1 \in (-\infty, \infty)$ $\alpha_2 > 0$

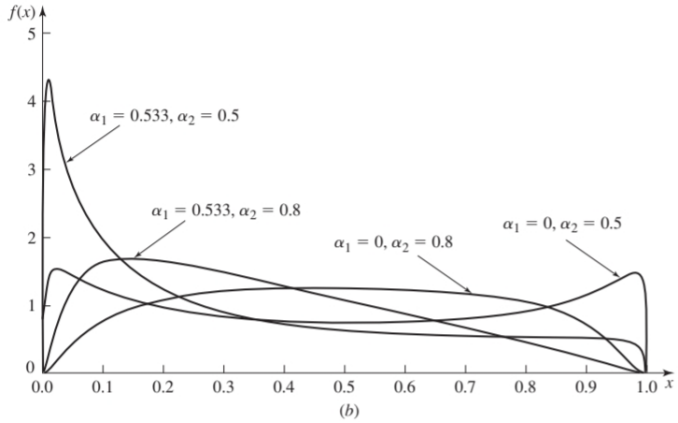
Función de densidad

$$f(x) = \begin{cases} \frac{\alpha_2(b-a)}{(x-a)(b-x)\sqrt{2\pi}} e^{-\frac{1}{2}[\alpha_1+\alpha_2 ln(\frac{x-a}{b-x})]} & \text{si } a < x < b \\ 0 & \text{otro caso} \end{cases}$$



Función acumulativa

$$F(x) = \begin{cases} \Phi[\alpha_1 + \alpha_2 ln(\frac{x-a}{b-x})] & \text{para } x > 0 \\ 0 & \text{otro caso} \end{cases}$$



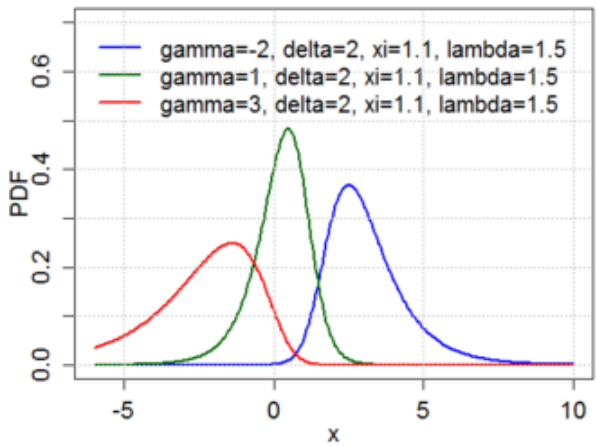
donde $\Phi(x)$ es la distribución Normal con $\mu = 0$ $\sigma^2 = 1$
No hay media y varianza establecida

Jhonson Unbounded $JSU(\alpha_1, \alpha_2, \gamma, \beta)$

Función de densidad

$$f(x) = \frac{\alpha_2}{\sqrt{2\pi}(x-\gamma)^2+\beta^2} e^{-\frac{1}{2}\left[\alpha_1+\alpha_2 ln\left[\frac{x-\gamma}{\beta}+\sqrt{\left(\frac{x-\gamma}{\beta})^2+1\right)}\right]\right]^2}$$

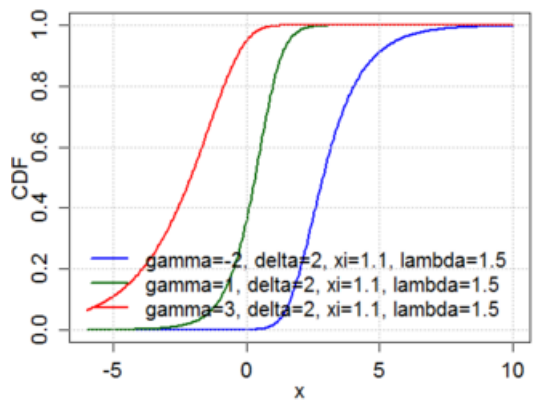
con $\infty < x < \infty$



Función acumulativa

$$F(x) = \Phi\left[\alpha_1 + \alpha_2 ln(\frac{x-\gamma}{\beta})] + \sqrt{\left(\frac{x-\gamma}{\beta})^2+1\right]}$$

con $-\infty < x < \infty$



$$E[X] = \gamma - \beta e^{\frac{1}{2\alpha_2^2}} senh\left(\frac{\alpha_1}{\alpha_2}\right)$$

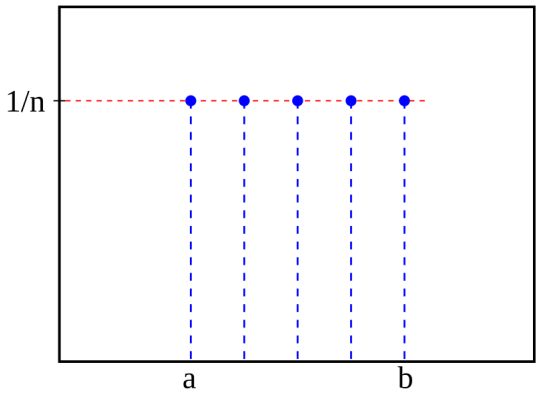
Distribuciones Discretas

$$E[X^n] = \sum_i x_i^n P(X = x_i)$$

Uniforme $U(n)$

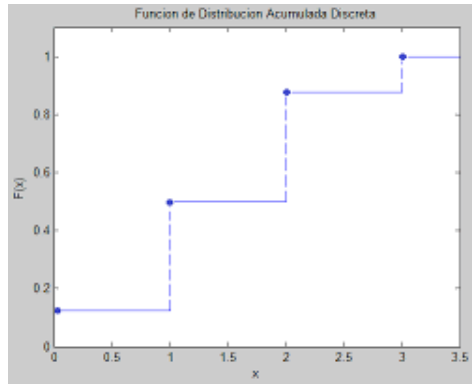
Función de probabilidad

$$f(x) = \frac{1}{n} \qquad x = 1, 2, \cdots, n$$



Función acumulativa

$$F(x) = \frac{x}{n} \qquad x = 1, 2, \cdots, n$$



$$E[x] = \frac{n+1}{2} \qquad V[x] = \frac{n^2-1}{12} \qquad FGM = \frac{e^{ta}(1-e^{tn})}{n(1-e^t)}$$

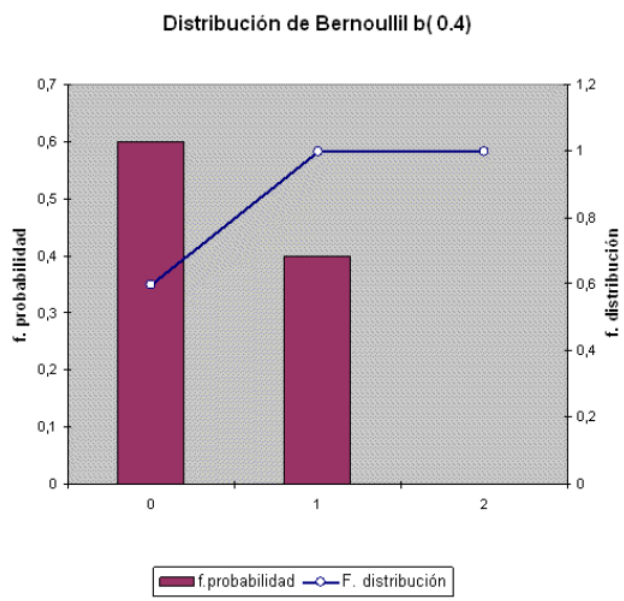
Bernoulli $B(p)$ $\quad p \in (0,1)$

Función de probabilidad

$$f(x) = \begin{cases} p & \text{si } x = 1, \\ 1-p & \text{si } x = 0, \\ 0 & \text{en otro caso.} \end{cases}$$

función acumulativa

$$F(x) = \begin{cases} 0 & \text{si } x < 0, \\ 1-p & \text{si } 0 \le x < 1, \\ 1 & \text{si } x \ge 1. \end{cases}$$



$$E[x] = p \qquad V[x] = p(1-p) \qquad \gamma_1 = \frac{q-p}{\sqrt{qp}} \qquad \gamma_2 = \frac{3p^2-3p+1}{pq} \qquad FGM = 1-p+pe^t$$

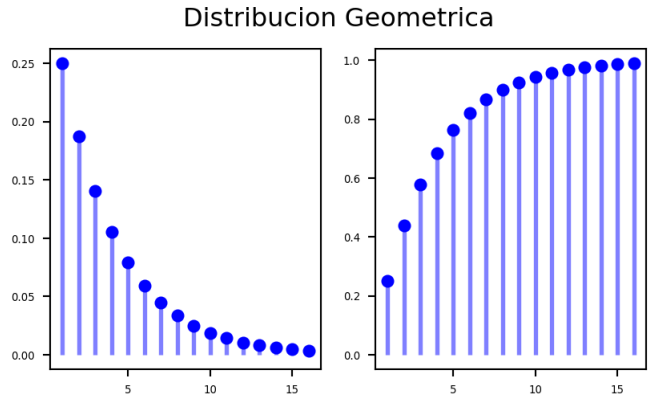
Geométrica $\quad geom(p) \quad p \in (0,1)$

Función de probabilidad

$$f(x) = \begin{cases} p(1-p)^x & \text{si } x = 0, 1, \cdots \\ 0 & \text{en otro caso.} \end{cases}$$

Función acumulativa

$$F(x) = \begin{cases} 0 & \text{otro caso,} \\ 1-(1-p)^{\lfloor x \rfloor + 1} & \text{si } x \geq 0. \end{cases}$$



$$E[x] = \frac{1}{p} \qquad V[x] = \frac{1-p}{p^2} \qquad \gamma_1 = \frac{2-p}{\sqrt{1-p}} \qquad \gamma_2 = 6 + \frac{p^2}{1-p} \qquad FGM = \frac{p}{1-(1-p)e^t}$$

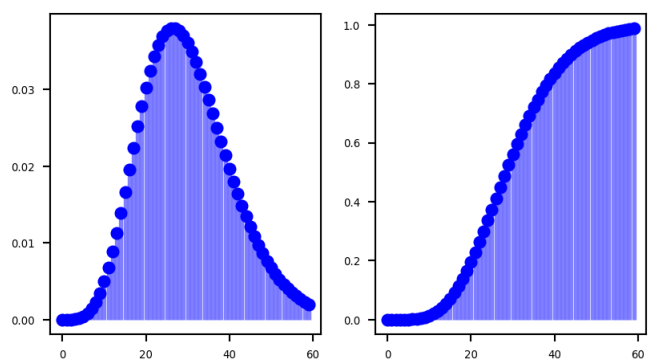
Binomial Negativa $\quad negbin(s,p) \quad s > 0 \quad p \in (0,1)$

Función de probabilidad

$$f(x) = \begin{cases} \binom{s+x-1}{x} p^s (1-p)^x & \text{si } x = 0, 1, 2, \cdot \\ 0 & \text{en otro caso.} \end{cases}$$

Función acumulativa

$$F(x) = \begin{cases} \sum_{i=0}^{\lfloor x \rfloor} \binom{s+i-1}{i} p^s (1-p)^i & \text{si } x \geq 0 \\ 0 & \text{en otro caso.} \end{cases}$$

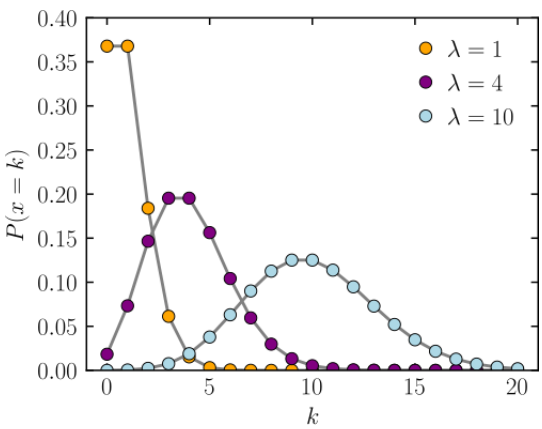


$$E[x] = \frac{s(1-p)}{p} \qquad V[x] = \frac{s(1-p)}{p^2} \qquad \gamma_1 = \frac{2-p}{\sqrt{s(1-p)}} \qquad \gamma_2 = \frac{6}{s} + \frac{p^2}{s(1-p)} \qquad FGM = \left(\frac{p}{1-(1-p)e^t}\right)^s$$

Poisson $p(\lambda)$ $\lambda > 0$ $x \in 0, 1, 2, \cdots$

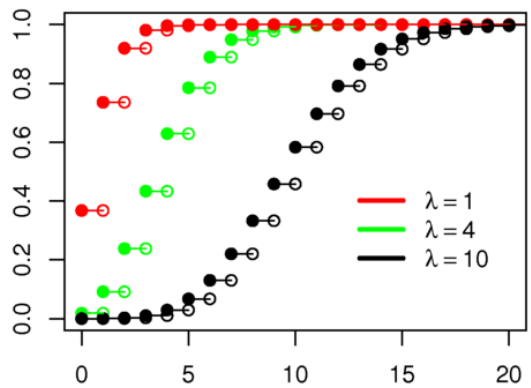
Función de probabilidad

$$f(x) = \begin{cases} \frac{e^{-\lambda}\lambda^x}{x!} & \text{si } x = 0, 1, 2, \cdot \\ 0 & \text{en otro caso.} \end{cases}$$



Función acumulativa

$$F(x) = \begin{cases} 0 & x < 0 \\ e^{-\lambda} \sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!} & \text{si } x \geq 0. \end{cases}$$



$$E[x] = \lambda = V[x] \qquad \gamma_1 = \lambda^{-1/2} \qquad \gamma_2 = 3 + \lambda^{-1} \qquad FGM = e^{\lambda(e^t - 1)}$$